

Mohammad Hasnain Raza

Los Angeles, CA

linkedin.com/in/hasnainraza03 | github.com/hasnainrazaa03

Email: razam@usc.edu

Mobile: (213) 994-5086

hasnainrazaa.vercel.app

EDUCATION

- **University of Southern California** Los Angeles, CA
M.S. in Computer Science; GPA: 3.67
◦ Coursework: Machine Learning, Operating Systems, Algorithms, Database Systems, Programming System Design
- **RV College of Engineering** Bengaluru, India
B.E. in Aerospace Engineering; GPA: 3.86, Silver Medalist
Aug 2018 – Jul 2022

EXPERIENCE

- **Sunbase Data** Orlando, FL (Remote)
AI Software Engineer Intern
May 2026 – Aug 2026
 - Fine-tuned **Ultralytics YOLO11** for multi-class roof-damage detection across 4 defect types on **5-7K annotated instances** drawn from ~2,000 aerial and ground-level roof images, and served it from a containerized **FastAPI** microservice.
 - Lifted the detector's **mAP@50 from 0.61 to 0.69** by mining shadows, vents and solar-panel edges back as **hard negatives**, after isolating them as the dominant false-positive sources via per-class precision/recall and confidence-threshold analysis.
 - Extended the team's GPT-4-class **sales agent** with **LangChain RAG** over **Pinecone**, adding conversation-history query rewriting and function-calling CRM tools for lead capture; validated on a **120-scenario** suite with out-of-scope cases.
- **Deloitte** Bengaluru, India
Technology Analyst
Aug 2022 – Nov 2024
 - Automated a **16-day** manual customer-onboarding process down to **2.5 days** in Pega PRPC, encoding tax, credit and legal approval chains as decision-tree routing over **7,500 customer creations and 12,500 modifications** across 35+ countries.
 - Validated the resulting **44% latency reduction** with a **Welch two-sample t-test** over **2,500+** before/after processing-time samples, at **p = 0.003** under sharply unequal pre- and post-automation variance.
 - Replaced manual intake triage with **few-shot LLM classification**, routing free-text requests into the 5 change types that determine the approval chain, at **92% agreement** with human annotators over an error-driven prompt-refinement loop.
- **Prana.ai** Remote
Founding Engineer (Part-time)
Sep 2019 – Dec 2021
 - Developed **SO(3)-equivariant CNNs** for volumetric super-resolution with the e3nn library, exploiting rotation-equivariant convolutions to lift reconstruction **SSIM ~35%** over trilinear interpolation, with PSNR agreeing.
 - Trained **depthwise-separable 3D CNNs** for tumor segmentation, reaching **<0.8s** GPU inference per volume through an architecture factorization cutting parameters 8-40× against a standard 3D stack.
 - Constructed end-to-end training pipelines over **5M+ MRI/CT images** from partner hospitals, applying intensity normalization, 64×64×64 patch extraction and class-balanced sampling against a 95:5 foreground imbalance.

PROJECTS

- **PeakRoutine – AI Health & Wellness Platform** Los Angeles, CA
Software Engineer | Claude, FastAPI, Spring Boot | #4 Product of the Day, Product Hunt
Mar 2026 – Present
 - Built the LLM layer of a health platform scoring **sleep, movement and nutrition** from wearable biometrics, shipping natural-language meal logging, **vision-based** photo recognition and six-persona coaching grounded in each user's own data.
 - Designed **7 acute pattern detectors and 8 chronic risk-trajectory models** that score against **rolling per-user baselines** rather than population thresholds, flagging deviations with Z-score anomaly and **CUSUM** change-point detection.
 - Engineered a **7-provider wearable ingestion platform** feeding those models, with Svix HMAC-verified webhooks, event deduplication and idempotent upserts, then owned the **Android bring-up** to Health Connect and Samsung Health.
- **Project Vimaan – AI Voice Command NLU System** Los Angeles, CA
PyTorch, Hugging Face, DistilBERT, ONNX Runtime, X-Plane 12
Sep 2025 – Present
 - Architected a schema-driven **synthetic data pipeline** producing **70,000+** label-verified voice commands across 6 slot types and **17 intent classes** (13 actionable, 3 conversational, 1 out-of-scope reject), with Pegasus and FLAN-T5 augmentation.
 - Fine-tuned a **joint intent-classification and BIO slot-tagging DistilBERT** to **98.9% accuracy, 0.967 macro-F1 and 0.99 slot F1 on a held-out test set**, after diagnosing and fixing **84.9% train/test leakage** in the evaluation harness.
 - Deployed inside **X-Plane 12** at **5.9ms** inference, **~9×** faster via sequence-length tuning and an **ONNX Runtime** backend, with INT8 quantization halving model size and **safety interlocks** gating irreversible commands behind voice confirmation.

SKILLS

- **Languages:** Python, Java, SQL, JavaScript, C/C++
- **ML & Deep Learning:** PyTorch, Hugging Face, YOLO11, DistilBERT, e3nn, TensorFlow, Scikit-learn, NumPy, Pandas
- **LLM & GenAI:** RAG, LangChain, Pinecone, Function Calling, Vision-Language Models, Prompt Engineering, Fine-tuning
- **CV & NLP:** Object Detection, 3D Segmentation, Super-Resolution, Intent Classification, Slot Filling, Sequence Labeling
- **Evaluation:** Held-out Evaluation, Leakage Detection, Error Analysis, Precision/Recall, mAP, Macro-F1, Ablation, Welch t-test
- **Serving & Data:** Docker, FastAPI, ONNX Runtime, Model Quantization, AWS, Async Pipelines, ETL, MongoDB